

Hyperlink Guide - Milestone 4: Core Development

Hello reviewer! We recognize that our documentation is substantial and complex. In order to help guide you, we provide here a series of links and explanations relating to the milestone outputs, acceptance criteria and evidence of milestone completion for Milestone 4.

While both the Outputs and Acceptance criteria include evidence and brief descriptions, each of our pieces of “evidence of milestone completion” includes summaries.

Please feel free to reach out if you have any questions.

Milestone Outputs

Comprehensive Internal Systems Testing:

- End-to-end functional testing of Sundial’s staking engine, bridge protocols (Bitcoin bridge integration/s), and DeFi smart contracts.
 - Staking & DeFi
 - <https://github.com/sundial-protocol/btc-locker> has
 - badges in README.md that show the latest status of the automated tests
 - tests within ‘/packages/core/tests’ that you can run with ‘npm run test’
 - a cli within ‘/packages/cli’ that can be installed locally with ‘npm run build’ and ‘npm run cli-link’ for manual testing
 - Bridge
 - <https://github.com/sundial-protocol/charms-cardano> has tests within the validator files. Run ‘aiken check’ to test locally.
- Identification and resolution of potential performance bottlenecks.
 - <https://github.com/sundial-protocol/sundial-monorepo/pull/57> bottleneck and general bug fixes listed within this repo

Security Audits:

- Third-party security audits assessing vulnerabilities in bridge protocols, staking mechanisms, and cross-chain transactions.
 - **Staking & Defi:**

Our staking & lending platform ‘btc-locker’ has been audited twice

 - Hacken Audit
 - <https://hacken.io/audits/sundial/dapp-sundial-btc-locker-apr2026/>

- Dual Defense
 - <https://hackenproof.com/audit-programs/sundial-web-and-dapp-du-aldefense-audit>
 - Charms-Cardano

We also secured a review of one of our bridge integrations

 - Josh Marchand
 - <https://github.com/yHSJ/charms-independent-review>
- Implementation of security best practices to protect against exploits and unauthorized access.
 - All audit remediations have been implemented and approved by audit teams.
 - We've also put together an overview of the best practices implemented in our work.

<https://github.com/sundial-protocol/internal-docs/blob/main/reports/M4.2-Best-Practices.md>

Scalability & Stress Testing:

- Load testing to evaluate Sundial's ability to handle high transaction volumes.
 - <https://github.com/sundial-protocol/sundial-monorepo/blob/main/demo/midgard-manager/packages/scalability-harness/benchmark-runs/30eb12b19e9b3e4a3dbf90ee93309d3a5aed7a3c/report.md> scalability report showcasing high load testing
- Network stress tests simulating peak demand to ensure stability under real-world conditions.
 - <https://github.com/sundial-protocol/sundial-monorepo/blob/main/demo/midgard-manager/packages/scalability-harness/benchmark-runs/30eb12b19e9b3e4a3dbf90ee93309d3a5aed7a3c/report.md> scalability report showcasing stability under large payload.

Acceptance criteria

Successful Functional Testing:

- Staking, bridge integration, and DeFi smart contracts execute transactions as expected in controlled test environments.
 - Staking & DeFi
 - <https://github.com/sundial-protocol/btc-locker> has:
 - badges in README.md that show the latest status of the automated tests
 - tests within '/packages/core/tests' that you can run with `npm run test`

- a cli within '/packages/cli' that can be installed locally with 'npm run build' and 'npm run cli-link' for manual testing
 - Bridge Integration
 - <https://github.com/sundial-protocol/charms-cardano> has tests within the validator files. Run `aiken check` to test locally.
- Bug tracking system shows critical issues resolved, with only minor bugs remaining before mainnet deployment. Documentation made public on Github and/or other platforms.
 - No critical issues remaining:
 - <https://github.com/sundial-protocol/sundial-monorepo/issues>

Security Audit Validation:

- No critical security vulnerabilities found; all major concerns addressed with documented fixes.
 - Staking / Lending (BTC Locker)
 - Hacken: No critical or major issues in the audits, minor remediations documented in the audit report:
 - <https://hacken.io/audits/sundial/dapp-sundial-btc-locker-apr2026/>
 - Dual Defense:
 - <https://hackenproof.com/audit-programs/sundial-web-and-dapp-dualdefense-audit>
 - This is an ongoing community bug bounty program with independent auditors triaging the bugs to validate incoming reports. Currently, no vulnerabilities have been found.
 - Bridging (Charms-Cardano)
 - Josh Marchand: 1 major issue - identified and resolved
 - <https://github.com/yHSJ/charms-independent-review>
 - Resolving commit:
 - <https://github.com/CharmsDev/charms-cardano/commit/2bfa2e7eb67dc85ec0227ce5d5156e525993858f>
 - Other (minor / informational) fixes:
 - <https://github.com/CharmsDev/charms-cardano/commit/82264ef238e0945da67bf258292c173e53f67a21>
 - <https://github.com/CharmsDev/charms-cardano/commit/c52584ff3682b89ec93266fa96fafbe674a0f6b4>
 - <https://github.com/CharmsDev/charms-cardano/commit/770970a6542c1038cc7f549020d2f691ef155758>
 - Audit reports from independent security firms published for transparency.
 - All links included in this doc have been published publicly.
 - We have also been pushing them across social media for awareness.
 - <https://x.com/SundialProtocol/status/2056293707662467458>

- <https://x.com/SundialProtocol/status/2055324203033808956>
- <https://x.com/hackenclub/status/2054929303851139350>
- https://x.com/sde_laney/status/2055329269786812782
- https://x.com/_MrHunt_/status/2054918600587100515
- <https://x.com/SundialProtocol/status/2057388154911363445>
- <https://x.com/HackenProof/status/2057384009731043691>
- https://x.com/sde_laney/status/2060170093607473492
- Hacken.
 - <https://hacken.io/audits/sundial/dapp-sundial-btc-locker-apr2026/>
 - <https://www.sundialprotocol.com/resources/audit>
- Dual Defense
 - <https://hackenproof.com/audit-programs/sundial-web-and-dapp-dualdefense-audit>
- Josh Marchand
 - <https://github.com/yHSJ/charms-independent-review>

Proven Scalability & Stability:

- System withstands high transaction volumes without significant latency, failures, or degradation.
 - <https://github.com/sundial-protocol/sundial-monorepo/blob/main/demo/midgard-manager/packages/scalability-harness/benchmark-runs/30eb12b19e9b3e4a3dbf90ee93309d3a5aed7a3c/report.md> scalability report showing high transaction volumes are capable without major issues
 - The 'executed run' reports in this doc include fee information
- Transaction speeds and fees remain stable under stress tests, meeting predefined performance benchmarks.
 - <https://github.com/sundial-protocol/sundial-monorepo/blob/main/demo/midgard-manager/packages/scalability-harness/benchmark-runs/30eb12b19e9b3e4a3dbf90ee93309d3a5aed7a3c/report.md> scalability report showing transaction speeds under predetermined stress tests
 - The 'executed run' reports in this doc include fee information as well <https://github.com/sundial-protocol/sundial-monorepo/blob/main/demo/midgard-manager/packages/scalability-harness/benchmark-runs/30eb12b19e9b3e4a3dbf90ee93309d3a5aed7a3c/2026-05-27T11-15-21.633Z-baseline-100-800-replay/report.md>

Evidence of milestone completion

Testing Results:

- GitHub repository updates with automated test results, bug fixes, and performance improvements.
 - L2
 - <https://github.com/sundial-protocol/sundial-monorepo/commits/main>
 - Staking / DeFi
 - <https://github.com/sundial-protocol/btc-locker/commits/main/>
 - Bridge Integration
 - <https://github.com/sundial-protocol/charms-cardano/commits/main/>

Summary: The above includes the core open source components of the work we've done toward each of our objectives. The links go directly to the list of updates (commits) for each repository.

- Internal reports detailing the success of functional tests across staking, bridging, and DeFi features.
 - Report is here:
<https://github.com/sundial-protocol/internal-docs/blob/main/reports/M4.7-Testing-Results.md>
- Live demo or recorded video walkthrough showcasing the test environment and transaction executions.
 -  Dashboard Demo v4.mp4

Summary: A video demonstration of just how easy Bitcoin Defi is with our staking & lending platform.

Security Audit Reports:

- Publication of third-party security audit findings, with a summary of issues identified and solutions implemented.
 - Helper doc for navigating all audits:
<https://github.com/sundial-protocol/internal-docs/blob/main/reports/M4.5-Auditing-Overview.md>
 - Hacken.

- <https://hacken.io/audits/sundial/dapp-sundial-btc-locker-apr2026/>
 - <https://www.sundialprotocol.com/resources/audit>
- Dual Defense
 - <https://hackenproof.com/audit-programs/sundial-web-and-dapp-dualdefense-audit>
- Josh Marchand
 - <https://github.com/yHSJ/charms-independent-review>
- Public updates on Sundial’s blog or social media channels detailing the security enhancements made.
 - X post with summary & links:
 - <https://x.com/SundialProtocol/status/2055324203033808956>
 - Hacken & Sundial have published them on their websites.
 - <https://hacken.io/audits/sundial/dapp-sundial-btc-locker-apr2026/>
 - <https://www.sundialprotocol.com/resources/audit>
 - And a post for the X space to raise awareness for direct questions
 - <https://x.com/SundialProtocol/status/2056293707662467458>

Summary: Proof of public updates can be found above, both from our website, X page and Hacken’s own media sources. Sundial were very happy with the reception and interactions on each of these posts, including highly technical releases, proving the community was engaged and informed throughout the process.

- X Spaces or YouTube AMA session where security teams and Sundial developers discuss findings and improvements.
 - X Space Recording:
 - <https://x.com/SundialProtocol/status/2056660627586760944>

Summary: With over 200 listeners, Sundial’s and Hacken’s X Space allowed our developers and Hacken’s auditors to discuss in detail the audit findings, technical architecture and overall structure of the results.

Scalability & Stress Test Reports:

- Benchmarking reports detailing how Sundial performs under heavy network load.
 - <https://github.com/sundial-protocol/sundial-monorepo/blob/main/demo/midgard-manager/packages/scalability-harness/benchmark-runs/30eb12b19e9b3e4a3dbf90>

[ee93309d3a5aed7a3c/2026-05-29T06-51-47.271Z-institutional-1000-replay/report.md](https://github.com/sundial-protocol/sundial-monorepo/blob/main/demo/midgard-manager/packages/scalability-harness/benchmark-runs/ee93309d3a5aed7a3c/2026-05-29T06-51-47.271Z-institutional-1000-replay/report.md)

Summary: This scalability report shows Sundial's L2 performance under a 1000 tps workload, with no significant or medium issues arising.

- Side-by-side performance comparisons before and after optimisations, with transaction speeds and cost metrics.
 - <https://github.com/sundial-protocol/sundial-monorepo/blob/main/demo/midgard-manager/packages/scalability-harness/benchmark-runs/e0b938c463b081b4cb7e3a7ab2a0634dc8efe689/report.md> scalability report before fixes were implemented
 - <https://github.com/sundial-protocol/sundial-monorepo/blob/main/demo/midgard-manager/packages/scalability-harness/benchmark-runs/30eb12b19e9b3e4a3dbf90ee93309d3a5aed7a3c/report.md> scalability report after fixes implemented

Summary: These documents include specific sub-reports highlighting tests of transaction speeds and associated costs for the L2 pre- and post-improvements. For ease, we have highlighted the exact sub-reports below:

Pre-improvement report:

<https://github.com/sundial-protocol/sundial-monorepo/blob/main/demo/midgard-manager/packages/scalability-harness/benchmark-runs/e0b938c463b081b4cb7e3a7ab2a0634dc8efe689/2026-05-21T13-16-18.076Z-initial-800-replay/report.md>

Post-improvement report:

<https://github.com/sundial-protocol/sundial-monorepo/blob/main/demo/midgard-manager/packages/scalability-harness/benchmark-runs/30eb12b19e9b3e4a3dbf90ee93309d3a5aed7a3c/2026-05-29T06-10-13.462Z-initial-800-replay/report.md>

- Community-accessible dashboards or graphs showing testnet transaction volumes and system performance.

Link to dashboard:

<http://dashboard.testnet.sundialprotocol.com/d/ee75b6b8-f1c6-4ef1-9d39-fe50cc55a274/midgard>

The Sundial L2 node ships a pre-configured Grafana dashboard providing real-time visibility into testnet transaction volumes and system performance. It auto-refreshes every 10 seconds and is available to anyone with access to the running node stack.

Transaction Volume

The dashboard tracks the full lifecycle of a transaction from submission to confirmed settlement on L1:

- *Submission throughput - how many L2 transactions the node is receiving per second, and the current depth of the processing queue*
- *Mempool depth - how many validated transactions are waiting to be batched into the next block*
- *Block fill rate - transaction count and total payload size per committed block, showing how densely each batch is packed*
- *L1 user events per block - count of authenticated on-chain events (deposits, withdrawals, transaction orders) included in each commitment*

Block Lifecycle Progress

Separate counters and rates expose each stage of the block pipeline:

- *Blocks built by the sequencer*
- *Blocks submitted to L1 as commitment transactions*
- *Blocks merged into the confirmed on-chain state queue*

Tracking all three stages makes it easy to see whether the pipeline is advancing end-to-end or stalling at a particular step.

Performance

The block commitment worker duration histogram shows how long the node takes to build and sign each block, giving a direct measure of sequencer latency under load.

Health Signals

Four panels serve as health indicators that should read zero during normal operation:

- *Block commitment failures - worker crashes or signing errors*
- *Merge failures - errors advancing the confirmed state on L1*
- *Rejected submissions - malformed CBOR rejected at the API boundary*
- *Submit 503 cause rate - backpressure rejections split by cause (queue full vs. offer timeout), useful for identifying overload*

Infrastructure

Container-level CPU, memory, and network I/O panels give a baseline view of node resource consumption alongside the application metrics, so performance degradation can be correlated with infrastructure pressure.